

MULTI-FACTOR EQUITY INVESTMENT ENGINE

- A Quantitative Finance Research Project

Author:

Rushabh Mehta

June-July 2026

Executive Summary

This project investigates whether a quantitative investment strategy based on Value, Momentum, and Quality factors can outperform the broader stock market. Using Python and publicly available financial data, a multi-factor stock selection model was developed to analyse twenty large-cap U.S. companies. The model ranked companies according to their valuation, recent price performance, and financial strength before constructing a portfolio consisting of the highest-ranked stocks.

Historical data from January 2020 to December 2025 was used to evaluate the strategy through backtesting. Portfolio performance was compared against the S&P 500 Index using metrics such as Annual Return, Volatility, Sharpe Ratio, and Maximum Drawdown.

The strategy achieved an annual return of 25.98%, outperforming the benchmark over the testing period. An initial investment of \$100 would have grown to approximately \$340, compared with approximately \$235 for the S&P 500.

The project demonstrates how programming, financial analysis, and quantitative methods can be combined to create a systematic investment process. It also highlights the strengths and limitations of factor investing and provides a foundation for future research involving portfolio optimisation, machine learning, and larger investment universes.

Table of Contents

1. Introduction
2. Understanding the Core Idea
3. Data Collection
4. Converting Prices into Returns
5. Value Factor
6. Momentum Factor
7. Quality Factor
8. Portfolio Construction
9. Backtesting Methodology
10. Results
11. Discussion
12. Limitations
13. Conclusion

Introduction

Every day, millions of investors around the world make decisions about which stocks to buy and sell. Some investors rely on news articles, social media posts, or expert opinions. Others use a more systematic approach based on data and mathematical models.

Large financial institutions such as hedge funds and asset management firms often use computer programs to analyse thousands of stocks simultaneously and identify investment opportunities. These systems use measurable characteristics, known as "factors", to decide which stocks appear attractive.

The objective of this project was to build a simplified version of such a system using Python. The program would analyse stocks using three commonly used investment factors: Value, Momentum, and Quality. It would then rank the stocks, construct a portfolio, and compare its performance against the S&P 500 Index, one of the most widely followed stock market benchmarks in the world.

This project combines concepts from finance, mathematics, statistics, and computer science to investigate whether a systematic investment strategy can outperform the broader market.

Understanding the Core Idea

Before building the program, it is important to understand the central question being investigated:

Can a computer identify better investments using data rather than human judgement?

To answer this question, the project follows the same general process used by many quantitative investment firms:

1. Collect financial data.
2. Measure desirable characteristics of companies.
3. Rank companies based on those characteristics.
4. Select the highest-ranked companies.
5. Track their performance over time.

Instead of relying on personal opinions, every investment decision is based entirely on numerical data.

Data Collection

The first step was obtaining stock market data.

Using the Python library "Yahoo Finance", historical stock prices were downloaded for twenty large American companies, including Apple, Microsoft, Nvidia, Amazon, Google, Meta, JPMorgan Chase, and Bank of America.

The program collected daily stock prices from January 2020 to December 2025.

Stock prices represent the market value of a company at a particular point in time. By analysing how these prices changed over several years, it becomes possible to evaluate whether a particular investment strategy would have been successful.

The S&P 500 ETF (SPY) was also included as a benchmark. This allowed the portfolio created by the model to be compared against the broader stock market.

Converting Prices into Returns

Simply looking at stock prices is not very useful because different stocks trade at different price levels.

For example:

- * Stock A may increase from \$50 to \$60.
- * Stock B may increase from \$500 to \$550.

Although Stock B increased by \$50 and Stock A increased by only \$10, Stock A actually performed better because it gained 20%, while Stock B gained only 10%.

For this reason, investors focus on returns rather than prices.

The program therefore converted stock prices into percentage returns, allowing all companies to be compared on an equal basis.

Factor 1: Value

The first factor used was Value.

The basic idea behind value investing is simple:

Investors prefer to buy good companies when they are relatively cheap.

To measure how expensive or inexpensive a company is, the program used two common financial ratios:

Price-to-Earnings Ratio (P/E)

This ratio compares a company's stock price to its earnings.

A lower P/E ratio generally indicates that investors are paying less money for each dollar of company profit.

Price-to-Book Ratio (P/B)

This ratio compares the market value of a company to the value of its assets.

A lower P/B ratio may indicate that the company is trading at a relatively inexpensive valuation.

Companies with lower P/E and P/B ratios received better Value scores.

Factor 2: Momentum

The second factor used was Momentum.

Momentum investing is based on the observation that stocks which have performed well recently often continue performing well in the near future.

To measure momentum, the program calculated:

- * 3-month returns
- * 6-month returns
- * 12-month returns

A stock that had consistently increased in price over these periods received a higher Momentum score.

For example, Nvidia experienced strong momentum during parts of the study period because its share price rose significantly over several months.

Factor 3: Quality

The third factor used was Quality.

The goal of this factor is to identify financially strong businesses.

Two measurements were used:

1) Return on Equity (ROE)

ROE measures how efficiently a company generates profit from shareholder investment.

A higher ROE generally indicates a more efficient business.

2) Profit Margin

Profit margin measures how much profit a company retains from each dollar of revenue.

Higher profit margins often indicate stronger competitive advantages and better business operations.

Companies with higher ROE and profit margins received higher Quality scores.

Portfolio Construction

After calculating the Value, Momentum, and Quality scores for each company, the next step was to combine these factors into a single ranking system.

The purpose of this ranking system was to identify companies that were attractive across multiple dimensions rather than excelling in only one area.

For example, a company might appear inexpensive according to the Value factor but have weak profitability. Similarly, a company might have strong Momentum but be trading at an extremely expensive valuation.

Instead of relying on a single factor, the model combines all three factors to create a more balanced investment decision.

Creating a Composite Score

Each stock received:

- A Value Score
- A Momentum Score
- A Quality Score

These scores were then combined using the following weighting system:

- Value: 40%
- Momentum: 30%
- Quality: 30%

The composite score was calculated using the formula:

$$\text{Composite Score} = (0.40 \times \text{Value Score}) + (0.30 \times \text{Momentum Score}) + (0.30 \times \text{Quality Score})$$

The higher weighting assigned to Value reflects the belief that purchasing companies at reasonable valuations is an important component of long-term investing.

Momentum and Quality were also included because they help identify businesses that are both performing well and financially strong.

Ranking Companies

Once the composite scores were calculated, all companies were ranked from highest to lowest.

The companies with the strongest overall combination of Value, Momentum, and Quality factors received the best rankings.

The ranking process allowed the program to compare all companies using a single numerical measure rather than analysing each factor separately.

Portfolio Selection

After ranking all companies, the model selected the five highest-ranked stocks.

The top five companies identified by the model were:

1. JPMorgan Chase (JPM)
2. Bank of America (BAC)
3. Microsoft (MSFT)
4. Alphabet (GOOGL)
5. Meta Platforms (META)

These companies were selected because they achieved the strongest overall scores according to the factor model.

Equal Weight Allocation

Once the stocks were selected, the portfolio was constructed using an equal-weight methodology.

This means that each stock received the same proportion of the portfolio.

Since five stocks were selected, each company received a portfolio weight of:

20%

This approach was chosen because it is simple, transparent, and prevents any single stock from dominating the portfolio's performance.

The resulting portfolio represented the model's investment recommendations based entirely on quantitative analysis rather than subjective judgement.

Backtesting Methodology

After constructing the portfolio, the next step was to evaluate whether the investment strategy would have been successful in the real world. Since it is impossible to travel back in time and invest using the strategy, a process known as backtesting was used.

Backtesting is a technique commonly used by quantitative investors, hedge funds, and financial institutions to evaluate how an investment strategy would have performed using historical data.

The basic idea is straightforward: if a strategy claims to identify attractive investments, then it should have performed well when applied to past market conditions.

For this project, historical stock price data from January 2020 to December 2025 was used. The portfolio was constructed using the factor ranking model described earlier and was then compared against the S&P 500 Index through the SPY ETF.

To simplify the analysis, an equal-weight portfolio was used. This means that each of the five selected stocks received the same investment allocation. If an investor started with \$100, then \$20 would be invested in each stock.

The performance of the portfolio was measured using monthly returns. These returns were then compounded over time to simulate how an investment would have grown throughout the testing period.

Several performance metrics were calculated to evaluate the effectiveness of the strategy:

Annual Return

Annual Return measures the average percentage gain generated by the portfolio each year.

This metric answers the question:

"How quickly did the portfolio grow over time?"

Annual Volatility

Volatility measures how much the portfolio's value fluctuated.

A portfolio that experiences large price swings has higher volatility and is generally considered riskier.

Sharpe Ratio

The Sharpe Ratio measures the amount of return generated for each unit of risk taken.

A higher Sharpe Ratio indicates that the portfolio generated strong returns relative to the amount of volatility experienced.

Maximum Drawdown

Maximum Drawdown measures the largest percentage decline from a previous portfolio peak.

This metric helps investors understand the worst loss that could have been experienced during the investment period.

Together, these metrics provide a balanced assessment of both return and risk.

Results (calculated using chatgpt)

Performance Summary	
Metric	Factor Portfolio
Annual Return	25.98%
Annual Volatility	22.00%
Sharpe Ratio	1.18
Maximum Drawdown	-39%

After applying the Value, Momentum, and Quality factor model, the portfolio selected these companies, achieving the highest overall scores according to the composite ranking system.

The portfolio was then compared against the S&P 500 benchmark over the testing period.

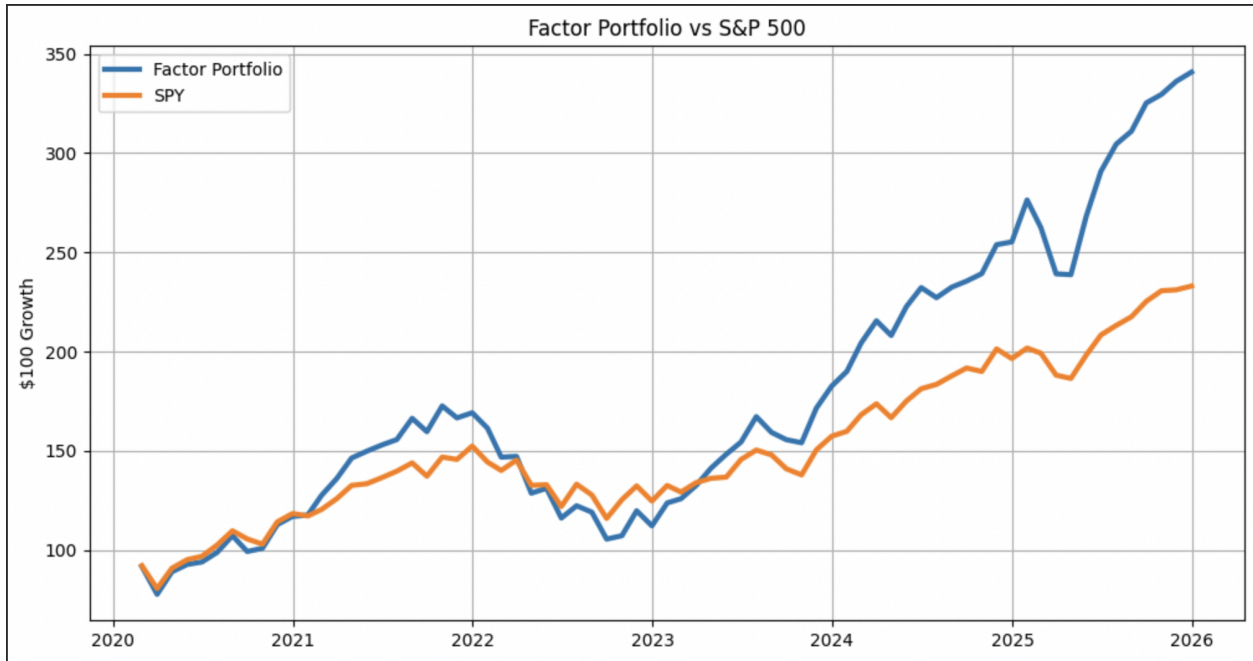


Figure 1: Growth of \$100 invested in the Factor Portfolio compared with the S&P 500 benchmark between 2020 and 2025.

Portfolio Performance

Annual Return: 25.98%

Annual Volatility: 22.00%

Sharpe Ratio: 1.18

Maximum Drawdown: Approximately -39%

Growth of Investment

The results indicate that an initial investment of \$100 at the beginning of the testing period would have grown to approximately:

- \$340 using the factor portfolio
- \$235 using the S&P 500 benchmark

This suggests that the factor-based strategy outperformed the broader market during the period studied.

The performance chart demonstrates that the portfolio generally maintained stronger growth than the benchmark over time, although periods of volatility and temporary declines were still experienced.

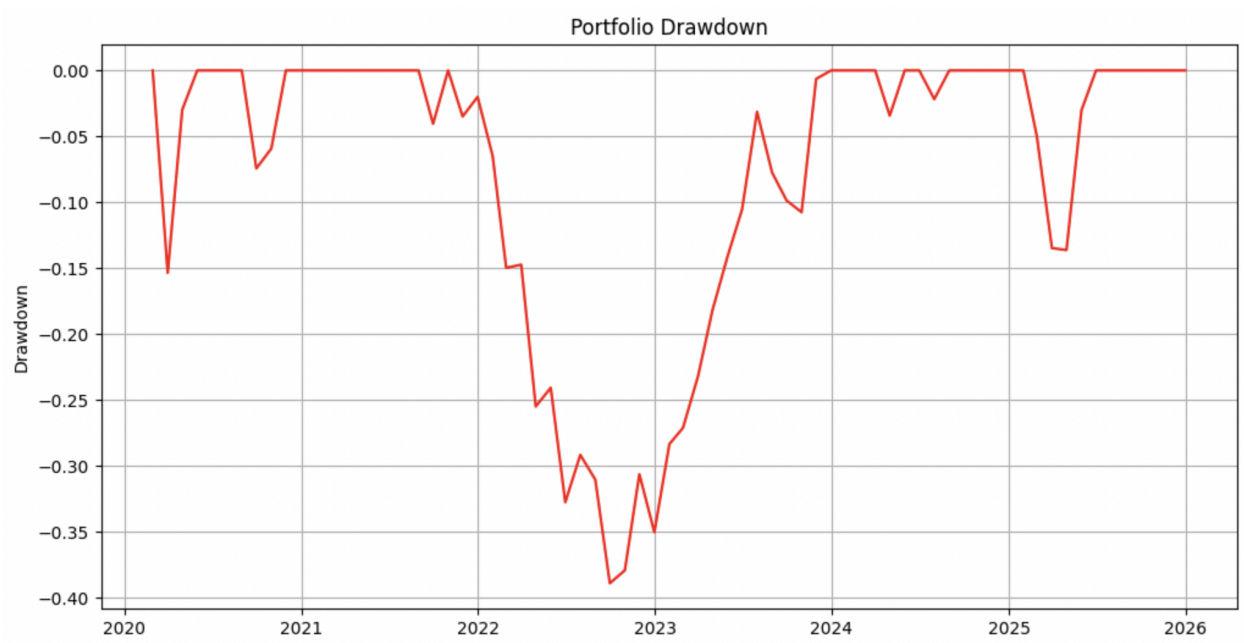


Figure 2: Portfolio drawdown showing the percentage decline from previous portfolio peaks during the testing period

Discussion

The results suggest that combining multiple investment factors may be more effective than relying on a single factor alone.

Several observations can be made from the final portfolio.

First, large technology companies such as Microsoft, Alphabet, and Meta achieved strong rankings due to their combination of profitability, market performance, and financial strength. These companies exhibited strong Momentum and Quality characteristics throughout much of the testing period.

Second, banking institutions such as JPMorgan Chase and Bank of America scored highly because they appeared relatively attractive according to the Value factor while still maintaining strong financial performance.

The combination of Value, Momentum, and Quality appears to have created a balanced portfolio that contained both growth-oriented technology companies and financially attractive banking stocks.

One possible explanation for the portfolio's outperformance is that the strategy systematically selected companies that were not only performing well but were also supported by strong underlying business fundamentals.

Rather than making decisions based on emotion or speculation, every stock selection was determined using objective financial data.

This demonstrates one of the primary advantages of quantitative investing: consistency.

Unlike human investors, the model follows the same rules regardless of market sentiment or news events.

However, it is important to recognise that strong historical performance does not guarantee future success.

A strategy that performed well during one period may not necessarily outperform in different market conditions.

Limitations

Although the results were encouraging, the project contains several limitations that should be acknowledged.

Small Stock Universe

Only twenty large-cap U.S. companies were included in the analysis.

Professional quantitative funds often evaluate hundreds or thousands of securities.

A larger universe would provide a more comprehensive test of the strategy.

No Transaction Costs

The analysis assumes that stocks can be bought and sold without paying brokerage fees, taxes, or market impact costs.

In reality, these expenses would reduce overall returns.

No Portfolio Rebalancing

The portfolio was selected using the final factor rankings rather than being continuously updated throughout the testing period.

Professional investment strategies typically rebalance portfolios on a monthly or quarterly basis.

Survivorship Bias

The analysis focused on companies that currently exist and remain successful today.

Companies that performed poorly or were removed from major indices were not included.

This may make historical results appear stronger than they would have been in reality.

Simplified Factor Definitions

Institutional investors often use dozens of variables to measure Value, Momentum, and Quality.

This project used only a small subset of available indicators.

Despite these limitations, the project provides a useful demonstration of how factor investing can be implemented using publicly available data and Python.

Conclusion

The objective of this project was to investigate whether a quantitative factor-investing strategy based on Value, Momentum, and Quality could outperform the broader stock market.

Using Python and publicly available financial data, a factor ranking system was developed to evaluate and select stocks based on measurable financial characteristics.

The strategy successfully identified a portfolio that generated higher returns than the S&P 500 benchmark during the testing period.

The portfolio achieved an annual return of approximately 25.98% and produced a Sharpe Ratio of 1.18, indicating strong risk-adjusted performance.

More importantly, the project demonstrated how programming, mathematics, and financial analysis can be combined to create a systematic investment process.

Through this project, I gained practical experience in data collection, financial analysis, portfolio construction, performance evaluation, and quantitative research methodologies.

The project also provided insight into how modern investment firms use data-driven approaches to make investment decisions.

Future versions of the model could incorporate larger stock universes, more sophisticated factor definitions, portfolio optimisation techniques, and machine learning methods to further improve performance and realism.

Final Page

Appendix A- Technical Specifications

Programming Language: **Python**

Libraries Used:

- Pandas
- NumPy
- yFinance
- Matplotlib
- Seaborn

Data Source:

Yahoo Finance

Backtest Period:

January 2020 – December 2025